Geetanjali Institute of Technical Studies

(Approved by AICTE, New Delhi and Affiliated to Rajasthan TechnicalUniversity Kota (Raj.))

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

B. Tech - VIII SEMESTER

तमसो मा ज्योतिर्गमय्

Darkness to Light

ACADEMIC YEAR – 2025-2026

Big Data Analytics Lab
8CS4-21

Submitted to:

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Submitted by:
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EXP NO: 1	Install Apache Hadoop
06/01/2026	

AIM: To Install Apache Hadoop.

Hadoop software can be installed in three modes of

Hadoop is a Java-based programming framework that supports the processing and storage of extremely large datasets on a cluster of inexpensive machines. It was the first major open source project in the big data playing field and is sponsored by the Apache Software Foundation.

Hadoop-2.7.3 is comprised of four main layers:

- **Hadoop Common** is the collection of utilities and libraries that support other Hadoop modules.
- **HDFS**, which stands for Hadoop Distributed File System, is responsible for persisting data to disk.
- **YARN**, short for Yet Another Resource Negotiator, is the "operating system" for HDFS.
- **MapReduce** is the original processing model for Hadoop clusters. It distributes work within the cluster or map, then organizes and reduces the results from the nodes into a response to a query. Many other processing models are available for the 2.x version of Hadoop.

Hadoop clusters are relatively complex to set up, so the project includes a stand-alone mode which is suitable for learning about Hadoop, performing simple operations, and debugging.

Procedure:

we'll install Hadoop in stand-alone mode and run one of the example example MapReduce programs it includes to verify the installation.

Prerequisites:

Step1: Installing Java 8 version.

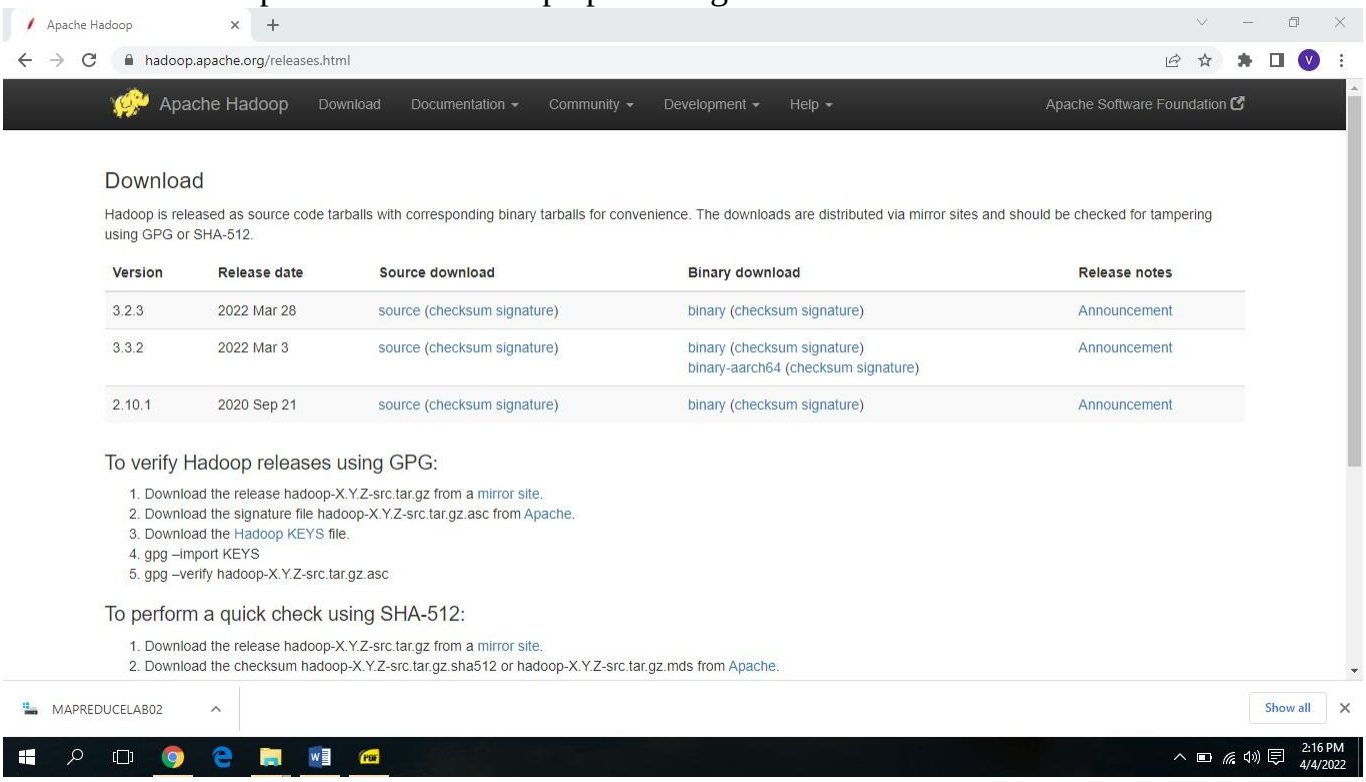
Openjdk version "1.8.0_91"
OpenJDK Runtime Environment (build 1.8.0_91-8u91-b14-3ubuntu1~16.04.1-b14)
OpenJDK 64-Bit Server VM (build 25.91-b14, mixed mode)
This output verifies that OpenJDK has been successfully installed.

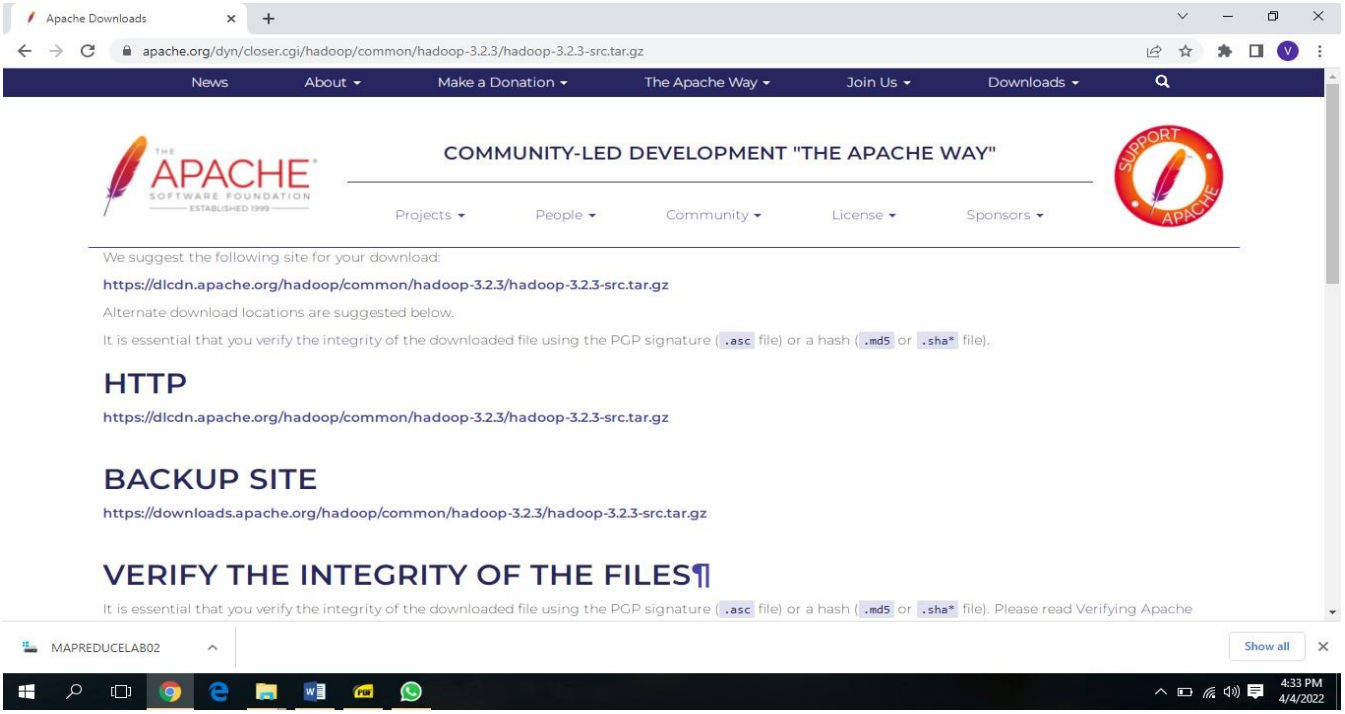
Note: To set the path for environment variables. i.e. JAVA_HOME

Step2: Installing Hadoop

With Java in place, we'll visit the Apache Hadoop Releases page to find the most recent stable release. Follow the binary for the current release:

Download Hadoop from www.hadoop.apache.org





Procedure to Run Hadoop

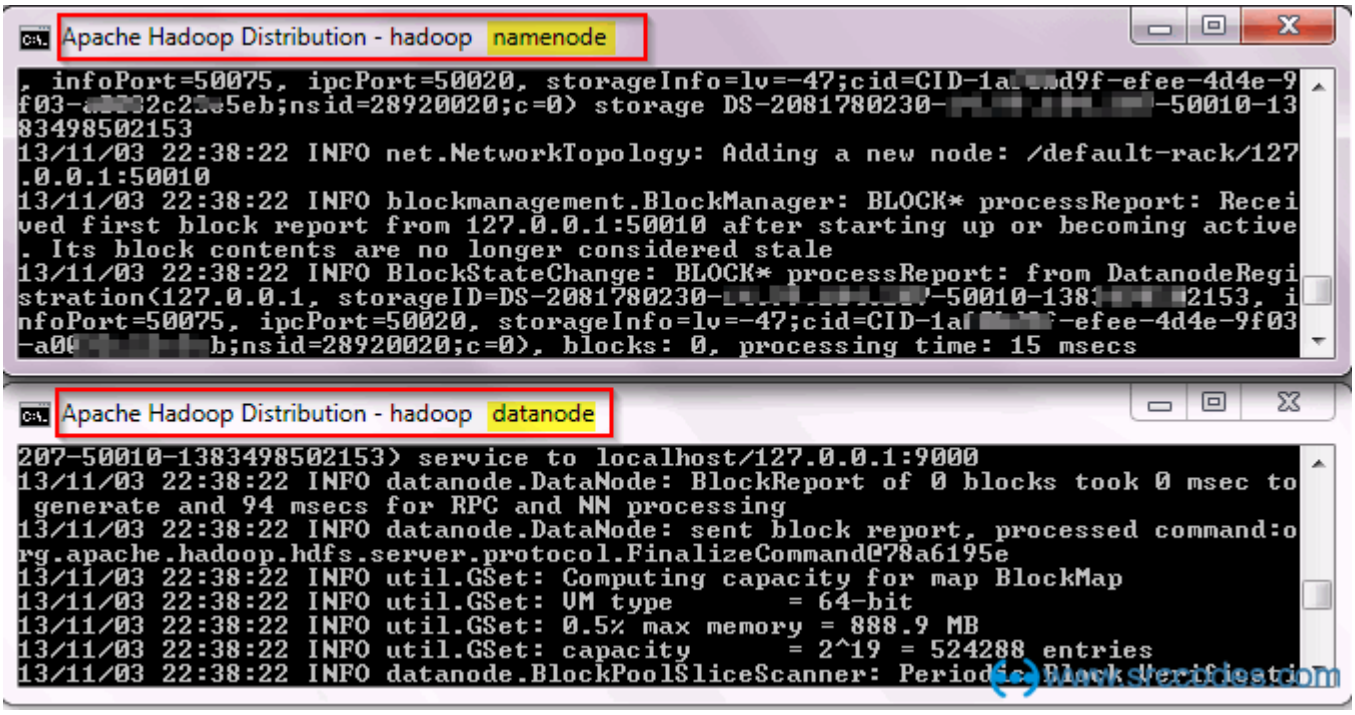
1. Install Apache Hadoop 2.2.0 in Microsoft Windows OS

If Apache Hadoop 2.2.0 is not already installed then follow the post Build, Install, Configure and Run Apache Hadoop 2.2.0 in Microsoft Windows OS.

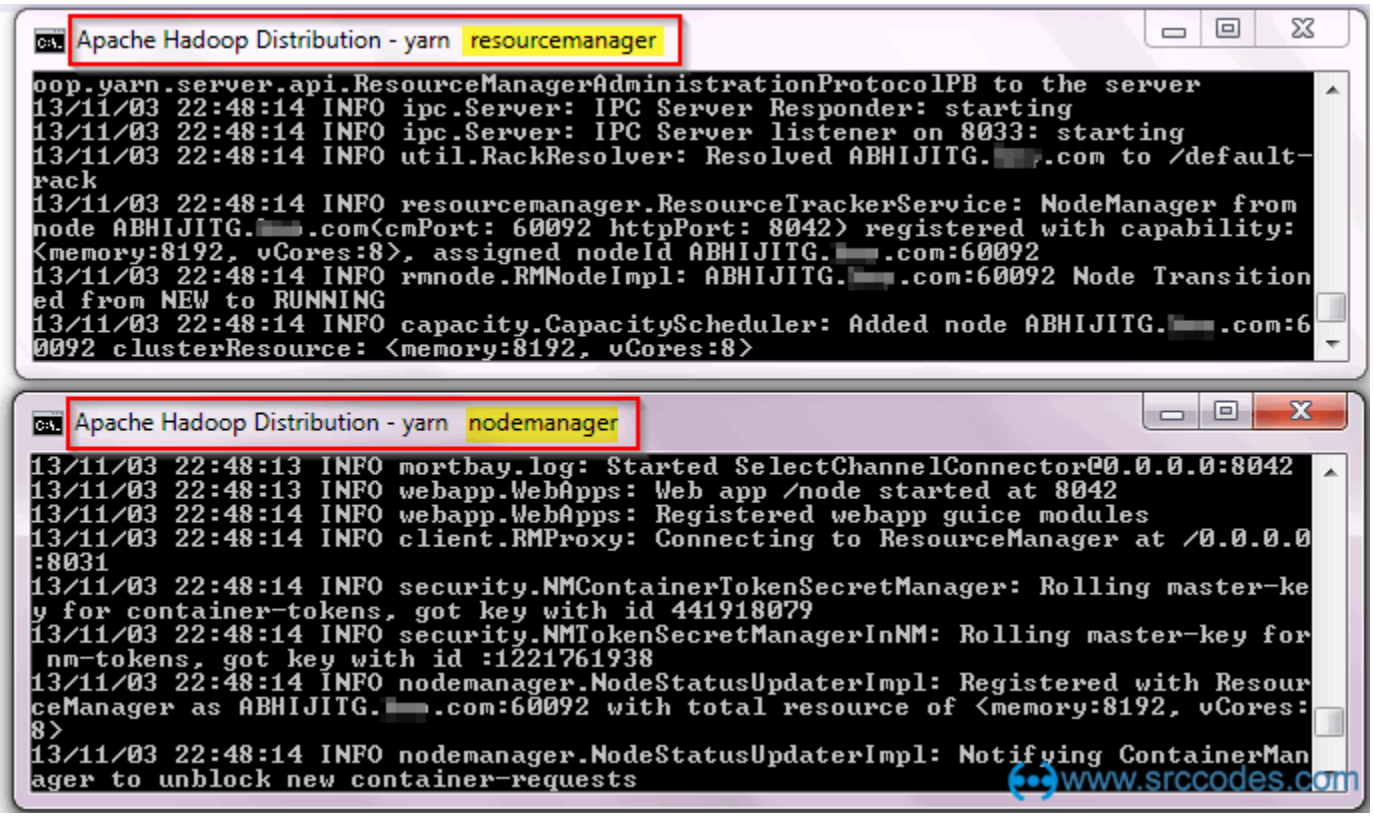
2. Start HDFS (Namenode and Datanode) and YARN (Resource Manager and Node Manager)

Run following commands.
Command Prompt
C:\Users\abhijitg>cd c:\hadoop
c:\hadoop>sbin\start-dfs
c:\hadoop>sbin\start-yarn
starting yarn daemons

Namenode, **Datanode**, **Resource Manager** and **Node Manager** will be started infew minutes and ready to execute Hadoop **MapReduce** job in the Single Node (pseudo-distributed mode) cluster.



Resource Manager & Node Manager:



Run wordcount MapReduce job

Now we'll run **wordcount** MapReduce job available in **%HADOOP_HOME%\share\hadoop\mapreduce\hadoop-mapreduce-examples-2.2.0.jar**

Create a text file with some content. We'll pass this file as input to the **wordcount** MapReduce job for counting words.
C:\file1.txt

```
Install  Hadoop
```

```
Run  Hadoop  Wordcount  Mapreduce  Example
```

Create a directory (say 'input') in HDFS to keep all the text files (say 'file1.txt') to be used for counting words.

C:\Users\abhijitg>cd c:\hadoop
C:\hadoop>bin\hdfs dfs -mkdir input

Copy the text file(say 'file1.txt') from local disk to the newly created 'input' directory in HDFS.

C:\hadoop>bin\hdfs dfs -copyFromLocal c:/file1.txt input

Check content of the copied file.

```
C:\hadoop>hdfs dfs -ls input
Found 1 items
-rw-r--r--   1 ABHIJITG supergroup         55 2014-02-03 13:19 input/file1.txt

C:\hadoop>bin\hdfs dfs -cat input/file1.txt
Install Hadoop
Run Hadoop Wordcount Mapreduce Example
```

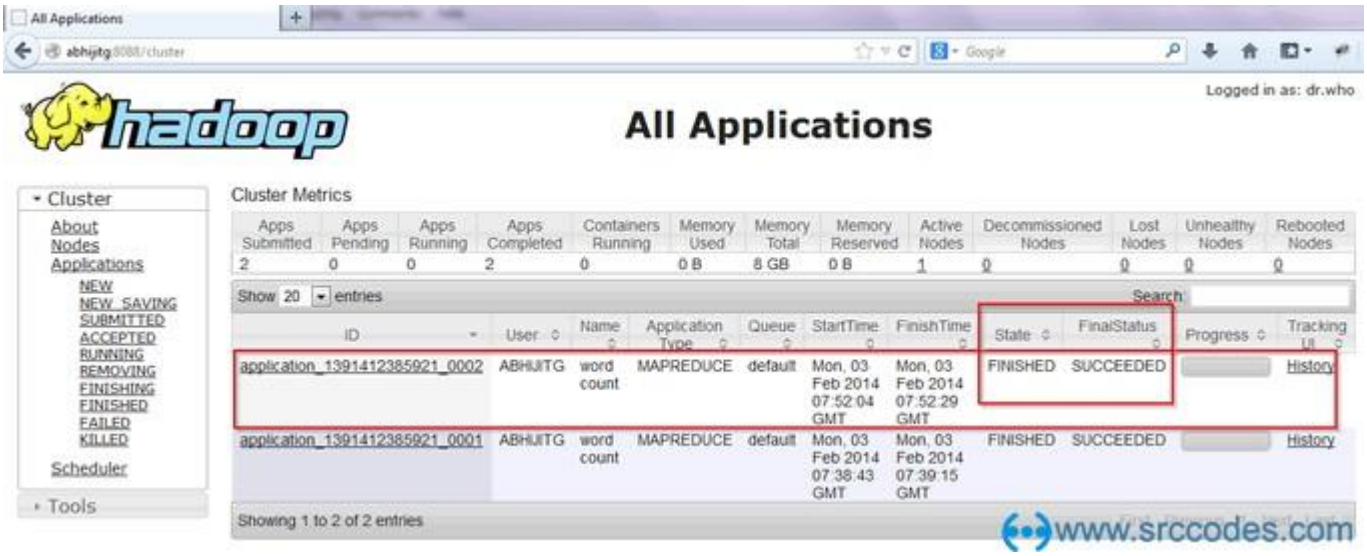
Run the wordcount MapReduce job provided in %HADOOP_HOME%\share\hadoop\mapreduce\hadoop-mapreduce-examples-2.2.0.jar

```
C:\hadoop>bin\yarn jar share/hadoop/mapreduce/hadoop-mapreduce-examples-
2.2.0.jar wordcount input output
14/02/03 13:22:02 INFO client.RMProxy: Connecting to ResourceManager at
/0.0.0.0:8032
14/02/03 13:22:03 INFO input.FileInputFormat: Total input paths to process : 1
14/02/03 13:22:03 INFO mapreduce.JobSubmitter: number of splits:1
:
:
14/02/03 13:22:04 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_1391412385921_0002
14/02/03 13:22:04 INFO impl.YarnClientImpl: Submitted application
application_1391412385921_0002 to ResourceManager at /0.0.0.0:8032
14/02/03 13:22:04 INFO mapreduce.Job: The url to track the job:
http://ABHIJITG:8088/proxy/application_1391412385921_0002/
14/02/03 13:22:04 INFO mapreduce.Job: Running job: job_1391412385921_0002
14/02/03 13:22:14 INFO mapreduce.Job: Job job_1391412385921_0002 running in
uber mode : false
14/02/03 13:22:14 INFO mapreduce.Job:  map 0% reduce 0%
14/02/03 13:22:22 INFO mapreduce.Job:  map 100% reduce 0%
14/02/03 13:22:30 INFO mapreduce.Job:  map 100% reduce 100%
14/02/03 13:22:30 INFO mapreduce.Job: Job job_1391412385921_0002 completed
successfully
14/02/03 13:22:31 INFO mapreduce.Job: Counters: 43
        File System Counters
                FILE: Number of bytes read=89
                FILE: Number of bytes written=160142
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
```

```
        HDFS: Number of bytes read=171
        HDFS: Number of bytes written=59
        HDFS: Number of read operations=6
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=2
    Job Counters
        Launched map tasks=1
        Launched reduce tasks=1
        Data-local map tasks=1
        Total time spent by all maps in occupied slots (ms)=5657
        Total time spent by all reduces in occupied slots (ms)=6128
    Map-Reduce Framework
        Map input records=2
        Map output records=7
        Map output bytes=82
        Map output materialized bytes=89
        Input split bytes=116
        Combine input records=7
        Combine output records=6
        Reduce input groups=6
        Reduce shuffle bytes=89
        Reduce input records=6
        Reduce output records=6
        Spilled Records=12
        Shuffled Maps =1
        Failed Shuffles=0
        Merged Map outputs=1
        GC time elapsed (ms)=145
        CPU time spent (ms)=1418
        Physical memory (bytes) snapshot=368246784
        Virtual memory (bytes) snapshot=513716224
        Total committed heap usage (bytes)=307757056
    Shuffle Errors
        BAD_ID=0
        CONNECTION=0
        IO_ERROR=0
        WRONG_LENGTH=0
        WRONG_MAP=0
        WRONG_REDUCE=0
    File Input Format Counters
        Bytes Read=55
    File Output Format Counters
```

Bytes Written=59

http://abhijitg:8088/cluster



Result: We've installed Hadoop in stand-alone mode and verified it by running an example program it provided.

EXP NO: 2	MapReduce program to calculate the frequency
13/01/2026	

AIM: To Develop a MapReduce program to calculate the frequency of a given word in agiven file

Map Function – It takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (Key-Value pair).

Example – (Map function in Word Count)

Input

Set of data

Bus, Car, bus, car, train, car, bus, car, train, bus, TRAIN,BUS, buS, caR, CAR, car, BUS, TRAIN

Output

Convert into another set of data

(Key,Value)

(Bus,1), (Car,1), (bus,1), (car,1), (train,1), (car,1), (bus,1), (car,1), (train,1), (bus,1), (TRAIN,1),(BUS,1), (buS,1), (caR,1), (CAR,1), (car,1), (BUS,1), (TRAIN,1)

Reduce Function – Takes the output from Map as an input and combines those data tuples into a smaller set of tuples.

Example – (Reduce function in Word Count)

Input Set of Tuples

(output of Map function)

(Bus,1), (Car,1), (bus,1), (car,1), (train,1), (car,1), (bus,1), (car,1), (train,1), (bus,1), (TRAIN,1),(BUS,1),

(buS,1),(caR,1),(CAR,1), (car,1), (BUS,1), (TRAIN,1)

Output **Converts into smaller set of tuples**

(BUS,7), (CAR,7), (TRAIN,4)

Work Flow of Program

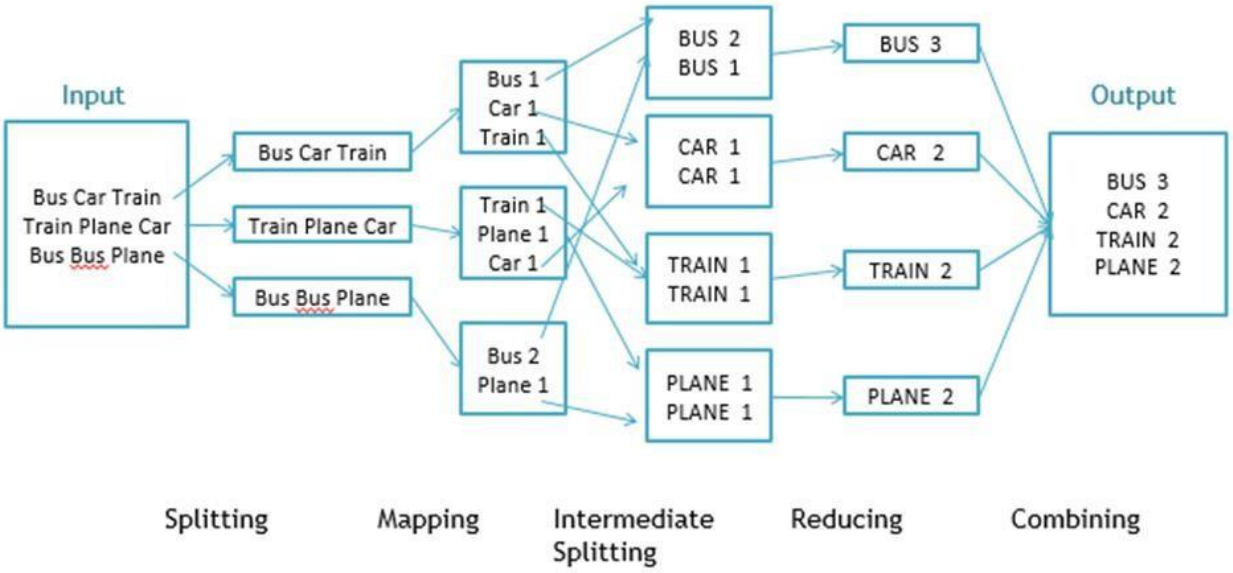


Fig. WorkFlow of MapReducing

Workflow of MapReduce consists of 5 steps

1. **Splitting –** The splitting parameter can be anything, e.g. splitting by space, comma, semicolon, or even by a new line ('\n').
2. **Mapping –** as explained above
3. Intermediate splitting – the entire process in parallel on different clusters. In order to group them in "Reduce Phase" the similar KEY data should be on same cluster.
4. **Reduce –** it is nothing but mostly group by phase
5. **Combining –** The last phase where all the data (individual result set from each cluster) is combine together to form a Result

Now Let's See the Word Count Program in Java

Make sure that Hadoop is installed on your system with java idk

Steps to follow

Step 1. Open Eclipse> File > New > Java Project > (Name it – MRProgramsDemo) > Finish
Step 2. Right Click > New > Package (Name it - PackageDemo) > Finish
Step 3. Right Click on Package > New > Class (Name it - WordCount)
Step 4. Add Following Reference Libraries –

Right Click on Project > Build Path> Add External Archivals

- /usr/lib/hadoop-0.20/hadoop-core.jar
- Usr/lib/hadoop-0.20/lib/Commons-cli-1.2.jar

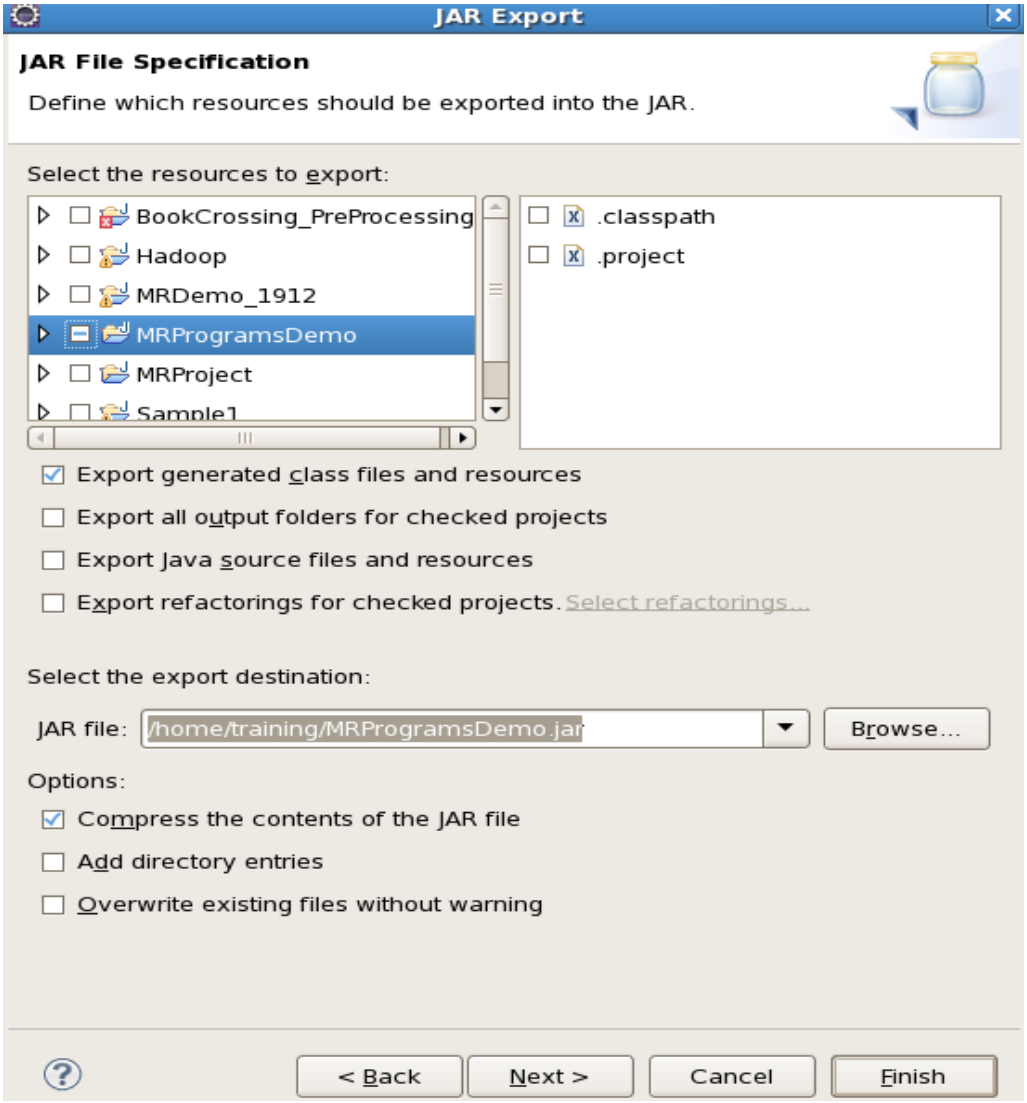
Program: Step 5. Type following Program :

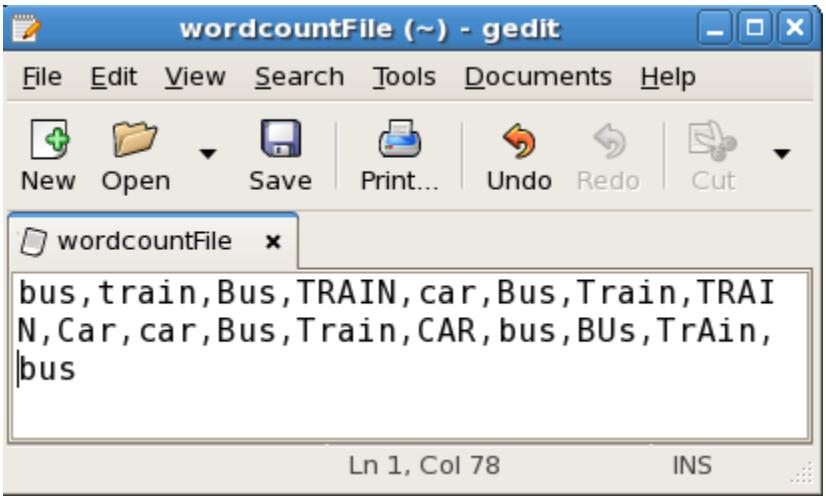
```
package PackageDemo;
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import org.apache.hadoop.util.GenericOptionsParser;
public class WordCount {
public static void main(String [] args) throws Exception
{
Configuration c=new Configuration();
String[] files=new GenericOptionsParser(c,args).getRemainingArgs();
Path input=new Path(files[0]);
Path output=new Path(files[1]);
Job j=new Job(c,"wordcount");
j.setJarByClass(WordCount.class);
j.setMapperClass(MapForWordCount.class);
j.setReducerClass(ReduceForWordCount.class);
j.setOutputKeyClass(Text.class);
j.setOutputValueClass(IntWritable.class);
FileInputFormat.addInputPath(j, input);
FileOutputFormat.setOutputPath(j, output);
System.exit(j.waitForCompletion(true)?0:1);
}
public static class MapForWordCount extends Mapper<LongWritable, Text, Text,
IntWritable>{
public void map(LongWritable  key, Text  value, Context  con) throws IOException,
InterruptedException
{
String line = value.toString();
```

```
String[] words=line.split(",");
for(String word: words )
{
   Text outputKey = new Text(word.toUpperCase().trim());
   IntWritable outputValue = new IntWritable(1);
   con.write(outputKey, outputValue);
}
}
}
public static class ReduceForWordCount extends Reducer<Text, IntWritable, Text,
IntWritable>
{
public void reduce(Text word, Iterable<IntWritable> values, Context con) throws
IOException,
InterruptedException
{
int sum = 0;
for(IntWritable value : values)
{
sum += value.get();
}
con.write(word, new IntWritable(sum));
}
}
}
```

Make Jar File

Right Click on Project> Export> Select export destination as Jar File > next> Finish


JAR Export
JAR File Specification
Define which resources should be exported into the JAR.
Select the resources to export:
BookCrossing_PreProcessing
Hadoop
MRDemo_1912
MRProgramsDemo
MRProject
Sample1
.classpath
.project
Export generated class files and resources
Export all output folders for checked projects
Export Java source files and resources
Export refactorings for checked projects. Select refactorings...
Select the export destination:
JAR file: /home/training/MRProgramsDemo.jar
Browse...
Options:
Compress the contents of the JAR file
Add directory entries
Overwrite existing files without warning
< Back
Next >
Cancel
Finish


wordcountFile (~) - gedit
File Edit View Search Tools Documents Help
New Open Save Print... Undo Redo Cut
wordcountFile
bus,train,Bus,TRAIN,car,Bus,Train,TRAI
N,Car,car,Bus,Train,CAR,bus,BUs,TrAin,
bus
Ln 1, Col 78
INS

To Move this into Hadoop directly, open the terminal and enter the following commands:
[training@localhost ~]$ hadoop fs -put wordcountFile wordCountFile

Run Jar file
(Hadoop jar jarfilename.jar packageName.ClassName PathToInputTextFile PathToOutputDirectry)

[training@localhost ~]$ Hadoop jar MRProgramsDemo.jar PackageDemo.WordCount wordCountFile MRDir1

Result: Open Result

```
[training@localhost ~]$ hadoop fs -ls MRDir1
Found 3 items
-rw-r--r-- 1 training supergroup
0 2016-02-23 03:36 /user/training/MRDir1/_SUCCESS
drwxr-xr-x - training supergroup
0 2016-02-23 03:36 /user/training/MRDir1/_logs
-rw-r--r-- 1 training supergroup
20 2016-02-23 03:36 /user/training/MRDir1/part-r-00000
[training@localhost ~]$ hadoop fs -cat MRDir1/part-r-00000
BUS     7
CAR    4
TRAIN 6
```

EXP NO: 3	MapReduce program to find the maximum temperature in each year
20/01/2026	

AIM: To Develop a MapReduce program to find the maximum temperature in each year.

Description: MapReduce is a programming model designed for processing large volumes of data in parallel by dividing the work into a set of independent tasks.Our previous traversal has given an introduction about MapReduce This traversal explains how to design a MapReduce program. The aim of the program is to find the Maximum temperature recorded for each year of NCDC data. The input for our program is weather data files for each year This weather data is collected by National Climatic Data Center – NCDC from weather sensors at all over the world. You can find weather data for each year from ftp://ftp.ncdc.noaa.gov/pub/data/noaa/.All files are zipped by year and the weather station. For each year, there are multiple files for different weather stations.
Here is an example for 1990 (ftp://ftp.ncdc.noaa.gov/pub/data/noaa/1901/).

- 010080-99999-1990.gz
- 010100-99999-1990.gz
- 010150-99999-1990.gz
- ……

MapReduce is based on set of key value pairs. So first we have to decide on the types for the key/value pairs for the input.
Map Phase: The input for Map phase is set of weather data files as shown in snap shot. The types of input key value pairs are LongWritable and Text and the types of output key value pairs are Text and IntWritable. Each Map task extracts the temperature data from the given year file. The output of the map phase is set of key value pairs. Set of keys are the years. Values are the temperature of each year.
Reduce Phase: Reduce phase takes all the values associated with a particular key. That is all the temperature values belong to a particular year is fed to a same reducer. Then each reducer finds the highest recorded temperature for each year. The types of output key value pairs in Map phase is same for the types of input key value pairs in reduce phase (Text and IntWritable). The types of output key value pairs in reduce phase is too Text and IntWritable.
So, in this example we write three java classes:

- HighestMapper.java
- HighestReducer.java
- HighestDriver.java

Program: HighestMapper.java

```
import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapred.*;
public class HighestMapper extends MapReduceBase implements Mapper<LongWritable, Text,
Text, IntWritable>
{
        public static final int MISSING = 9999;
public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output,
Reporter reporter) throws IOException
{
String line = value.toString();
String year = line.substring(15,19);
int temperature;
if (line.charAt(87)=='+')
temperature = Integer.parseInt(line.substring(88, 92));
else
temperature = Integer.parseInt(line.substring(87, 92));
String quality = line.substring(92, 93);
if(temperature != MISSING && quality.matches("[01459]"))
output.collect(new Text(year),new IntWritable(temperature));
}
}
```

HighestReducer.java

```
import java.io.IOException;
import  java.util.Iterator;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapred.*;
public class HighestReducer extends MapReduceBase implements Reducer<Text, IntWritable,
Text, IntWritable>
{
public void reduce(Text key, Iterator<IntWritable> values, OutputCollector<Text, IntWritable>
output, Reporter reporter) throws IOException
{
int max_temp = 0;
;
while (values.hasNext())
{
```

```
int current=values.next().get();
if ( max_temp < current)
max_temp = current;
}
output.collect(key, new IntWritable(max_temp/10));
}
```

HighestDriver.java

```
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.conf.*;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapred.*;
import org.apache.hadoop.util.*;
public class HighestDriver extends Configured implements Tool{
public int run(String[] args) throws Exception
{
JobConf conf = new JobConf(getConf(), HighestDriver.class);
conf.setJobName("HighestDriver");
conf.setOutputKeyClass(Text.class);
conf.setOutputValueClass(IntWritable.class);
conf.setMapperClass(HighestMapper.class);
conf.setReducerClass(HighestReducer.class);
Path inp = new Path(args[0]);
Path out = new Path(args[1]);
FileInputFormat.addInputPath(conf, inp);
FileOutputFormat.setOutputPath(conf, out);
JobClient.runJob(conf);
return 0;
}
public static void main(String[] args) throws Exception
{
int res = ToolRunner.run(new Configuration(), new HighestDriver(),args);
System.exit(res);
}
}
```

EXP NO: 4	MapReduce program to find the grades of student's Result
27/01/2026	

AIM: To Develop a MapReduce program to find the grades of student's**.**

```
import java.util.Scanner;
public class JavaExample
{
   public static void main(String args[])
   {
         /* This program assumes that the student has 6 subjects,
          * thats why I have created the array of size 6. You can
          * change this as per the requirement.
          */
      int marks[] = new int[6];
      int i;
      float total=0, avg;
      Scanner scanner = new Scanner(System.in);
      for(i=0; i<6; i++) {
        System.out.print("Enter Marks of Subject"+(i+1)+":");
        marks[i] = scanner.nextInt();
        total = total + marks[i];
      }
      scanner.close();
      //Calculating average
      here avg = total/6;
      System.out.print("The student Grade is: ");
      if(avg>=80)
      {
         System.out.print("A");
      }
      else if(avg>=60 && avg<80)
      {
        System.out.print("B");
      }
      else if(avg>=40 && avg<60)
      {
```

```
            System.out.print("C");
        }
             else
        {

            System.out.print("D");

        }

    }

}
```

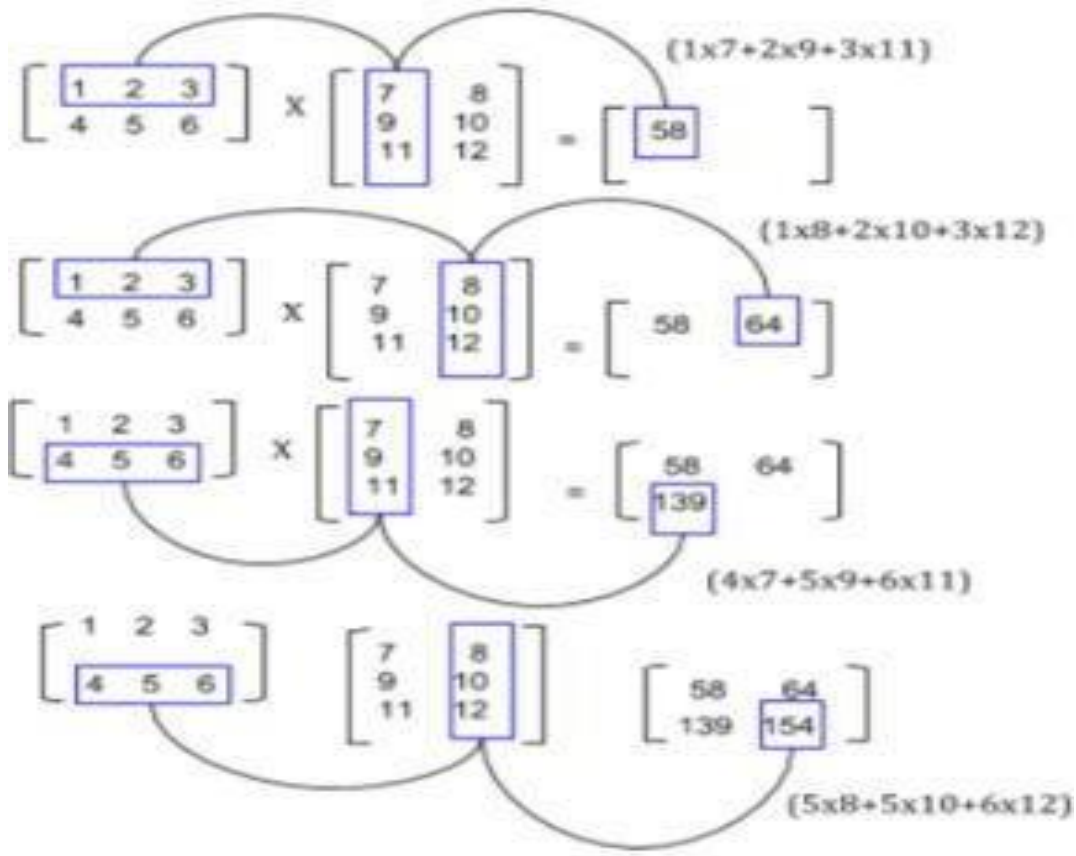
Expected Output:

```
Enter Marks of Subject1:40
Enter Marks of Subject2:80
Enter Marks of Subject3:80
Enter Marks of Subject4:40
Enter Marks of Subject5:60
Enter Marks of Subject6:60
The student Grade is: B
```

EXP NO: 5	MapReduce program to implement Matrix Multiplication
03/02/2026	

AIM: To Develop a MapReduce program to implement Matrix Multiplication.

In **mathematics**, **matrix multiplication** or the **matrix product** is a binary operation that produces a matrix from two matrices. The definition is motivated by linear equations and linear transformations on vectors, which have numerous applications in applied mathematics, physics, and engineering. In more detail, if **A** is an $n \times m$ matrix and **B** is an $m \times p$ matrix, their matrix product **AB** is an $n \times p$ matrix, in which the m entries across a row of **A** are multiplied with the m entries down a column of **B** and summed to produce an entry of **AB**. When two linear transformations are represented by matrices, then the matrix product represents the composition of the two transformations.



Algorithm for Map Function.

a. for each element m_{ij} of M do

produce (key,value) pairs as ((i,k), (M,j,m_{ij}), for k=1,2,3,.. upto the number of columns of N

b. for each element njk of N do
   produce (key,value) pairs as ((i,k),(N,j,N_{jk}), for i = 1,2,3,.. Upto the number of rows of M.

c. return Set of (key,value) pairs that each key (i,k), has list with values (M,j,m_{ij}) and (N, j,n_{jk}) for all possible values of j.

Algorithm for Reduce Function.

d. for each key (i,k) do
e. sort values begin with M by j in listM sort values begin with N by j in listN multiply mij and njk for jth value of each list
f. sum up mij x njk return (i,k), Σj=1 mij x njk

Step 1. Download the hadoop jar files with these links.

Download Hadoop Common Jar files: https://goo.gl/G4MyHp
$ wget https://goo.gl/G4MyHp -O hadoop-common-2.2.0.jar
Download Hadoop Mapreduce Jar File: https://goo.gl/KT8yfB
$ wget https://goo.gl/KT8yfB -O hadoop-mapreduce-client-core-2.7.1.jar

Step 2. Creating Mapper file for Matrix Multiplication.

```
import java.io.DataInput;
import java.io.DataOutput;
import java.io.IOException;
import java.util.ArrayList;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.io.Writable;
import org.apache.hadoop.io.WritableComparable;
import org.apache.hadoop.mapreduce.Job;
```

```
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.*;
import org.apache.hadoop.mapreduce.lib.output.*;
import org.apache.hadoop.util.ReflectionUtils;

class Element implements Writable {
	int tag;
	int index;
	double value;
	Element() {
		tag = 0;
		index = 0;
		value = 0.0;
	}
	Element(int tag, int index, double value) {
		this.tag = tag;
		this.index = index;
		this.value = value;
	}
	@Override
	public void readFields(DataInput input) throws IOException {
		tag = input.readInt();
		index = input.readInt();
		value = input.readDouble();
	}
	@Override
	public void write(DataOutput output) throws IOException {
		output.writeInt(tag);
		output.writeInt(index);
		output.writeDouble(value);
	}
}
class Pair implements WritableComparable<Pair> {
	int i;
	int j;

	Pair() {
		i = 0;
```

```
		j = 0;
	}
	Pair(int i, int j) {
		this.i = i;
		this.j = j;
	}
	@Override
	public void readFields(DataInput input) throws IOException {
		i = input.readInt();
		j = input.readInt();
	}
	@Override
	public void write(DataOutput output) throws IOException {
		output.writeInt(i);
		output.writeInt(j);
	}
	@Override
	public int compareTo(Pair compare) {
		if (i > compare.i) {
			return 1;
		} else if ( i < compare.i) {
			return -1;
		} else {
			if(j > compare.j) {
				return 1;
			} else if (j < compare.j) {
				return -1;
			}
		}
		return 0;
	}
	public String toString() {
		return i + " " + j + " ";
	}
}
public class Multiply {
	public static class MatriceMapperM extends Mapper<Object,Text,IntWritable,Element>
{
```

```
		@Override
		public void map(Object key, Text value, Context context)
				throws IOException, InterruptedException {
			String readLine = value.toString();
			String[] stringTokens = readLine.split(",");

			int index = Integer.parseInt(stringTokens[0]);
			double elementValue = Double.parseDouble(stringTokens[2]);
			Element e = new Element(0, index, elementValue);
			IntWritable keyValue = new
IntWritable(Integer.parseInt(stringTokens[1]));
			context.write(keyValue, e);
		}
	}
	public static class MatriceMapperN extends Mapper<Object,Text,IntWritable,Element> {
		@Override
		public void map(Object key, Text value, Context context)
				throws IOException, InterruptedException {
			String readLine = value.toString();
			String[] stringTokens = readLine.split(",");
			int index = Integer.parseInt(stringTokens[1]);
			double elementValue = Double.parseDouble(stringTokens[2]);
			Element e = new Element(1,index, elementValue);
			IntWritable keyValue = new
IntWritable(Integer.parseInt(stringTokens[0]));
			context.write(keyValue, e);
		}
	}
	public static class ReducerMxN extends Reducer<IntWritable,Element, Pair,
DoubleWritable> {
	@Override
	public void reduce(IntWritable key, Iterable<Element> values, Context context) throws
IOException, InterruptedException {
		ArrayList<Element> M = new ArrayList<Element>();
		ArrayList<Element> N = new ArrayList<Element>();
		Configuration conf = context.getConfiguration();
		for(Element element : values) {
			Element tempElement = ReflectionUtils.newInstance(Element.class,
conf);
```

```
                ReflectionUtils.copy(conf, element, tempElement);

                if (tempElement.tag == 0) {
                        M.add(tempElement);
                } else if(tempElement.tag == 1) {
                        N.add(tempElement);
                }
        }
        for(int i=0;i<M.size();i++) {
                for(int j=0;j<N.size();j++) {

                        Pair p = new Pair(M.get(i).index,N.get(j).index);
                        double multiplyOutput = M.get(i).value * N.get(j).value;

                        context.write(p, new DoubleWritable(multiplyOutput));
                        }
                }
        }
    }
    public static class MapMxN extends Mapper<Object, Text, Pair, DoubleWritable> {
        @Override
        public void map(Object key, Text value, Context context)
                        throws IOException, InterruptedException {
                String readLine = value.toString();
                String[] pairValue = readLine.split(" ");
                Pair p = new
Pair(Integer.parseInt(pairValue[0]),Integer.parseInt(pairValue[1]));
                DoubleWritable val = new
DoubleWritable(Double.parseDouble(pairValue[2]));
                context.write(p, val);
        }
    }
    public static class ReduceMxN extends Reducer<Pair, DoubleWritable, Pair,
DoubleWritable> {
        @Override
        public void reduce(Pair key, Iterable<DoubleWritable> values, Context context)
throws IOException, InterruptedException {
                double sum = 0.0;
                for(DoubleWritable value : values) {
```

```
                sum += value.get();
            }
            context.write(key, new DoubleWritable(sum));
        }
    }
    public static void main(String[] args) throws Exception {
        Job job = Job.getInstance();
        job.setJobName("MapIntermediate");
        job.setJarByClass(Project1.class);
        MultipleInputs.addInputPath(job, new Path(args[0]), TextInputFormat.class,
MatriceMapperM.class);
        MultipleInputs.addInputPath(job, new Path(args[1]), TextInputFormat.class,
MatriceMapperN.class);
        job.setReducerClass(ReducerMxN.class);
        job.setMapOutputKeyClass(IntWritable.class);
        job.setMapOutputValueClass(Element.class);
        job.setOutputKeyClass(Pair.class);
        job.setOutputValueClass(DoubleWritable.class);
        job.setOutputFormatClass(TextOutputFormat.class);
        FileOutputFormat.setOutputPath(job, new Path(args[2]));
        job.waitForCompletion(true);
        Job job2 = Job.getInstance();
        job2.setJobName("MapFinalOutput");
        job2.setJarByClass(Project1.class);

        job2.setMapperClass(MapMxN.class);
        job2.setReducerClass(ReduceMxN.class);

        job2.setMapOutputKeyClass(Pair.class);
        job2.setMapOutputValueClass(DoubleWritable.class);

        job2.setOutputKeyClass(Pair.class);
        job2.setOutputValueClass(DoubleWritable.class);

        job2.setInputFormatClass(TextInputFormat.class);
        job2.setOutputFormatClass(TextOutputFormat.class);

        FileInputFormat.setInputPaths(job2, new Path(args[2]));
        FileOutputFormat.setOutputPath(job2, new Path(args[3]));
```

```
                job2.waitForCompletion(true);
        }
}
```

Step 5. Compiling the program in particular folder named as operation

```
#!/bin/bash

rm -rf multiply.jar classes

module load hadoop/2.6.0

mkdir -p classes
javac -d classes -cp classes:`$HADOOP_HOME/bin/hadoop classpath` Multiply.java
jar cf multiply.jar -C classes .

echo "end"
```

Step 6. Running the program in particular folder named as operation

```
export HADOOP_CONF_DIR=/home/$USER/cometcluster
module load hadoop/2.6.0
myhadoop-configure.sh
start-dfs.sh
start-yarn.sh

hdfs dfs -mkdir -p /user/$USER
hdfs dfs -put M-matrix-large.txt /user/$USER/M-matrix-large.txt
hdfs dfs -put N-matrix-large.txt /user/$USER/N-matrix-large.txt
hadoop jar multiply.jar edu.uta.cse6331.Multiply /user/$USER/M-matrix-large.txt
/user/$USER/N-matrix-large.txt /user/$USER/intermediate /user/$USER/output
rm -rf output-distr
mkdir output-distr
hdfs dfs -get /user/$USER/output/part* output-distr

stop-yarn.sh
stop-dfs.sh
myhadoop-cleanup.sh
```

EXP NO: 6	MapReduce to find the maximum electrical consumption in each year
10/02/2026	

AIM: To Develop a MapReduce to find the maximum electrical consumption in each year given electrical consumption for each month in each year.

Given below is the data regarding the electrical consumption of an organization. It contains the monthly electrical consumption and the annual average for various years.

If the above data is given as input, we have to write applications to process it and produce results such as finding the year of maximum usage, year of minimum usage, and so on. This is a walkover for the programmers with finite number of records. They will simply write the logic to produce the required output, and pass the data to the application written.

But, think of the data representing the electrical consumption of all the largescale industries of a particular state, since its formation.

When we write applications to process such bulk data,

- They will take a lot of time to execute.
- There will be a heavy network traffic when we move data from source to network server and so on.

To solve these problems, we have the MapReduce framework

Input Data

The above data is saved as sample.txt and given as input. The input file looks as shown below.

```
1979   23  23   2      43      24      25      26      26      26      26      25      26      25
1980   26  27  28      28      28      30      31      31      31      30      30      30      29
1981   31  32  32      32      33      34      35      36      36      34      34      34      34
1984   39  38  39      39      39      41      42      43      40      39      38      38      40
1985   38  39   39   39 39   41  41   41  00   40  39   39 45
```

Source code:

```
import java.util.*;
import java.io.IOException;
import java.io.IOException;
import org.apache.hadoop.fs.Path;
import  org.apache.hadoop.conf.*;
import org.apache.hadoop.io.*;
```

```
import org.apache.hadoop.mapred.*;
import org.apache.hadoop.util.*;
public class ProcessUnits
{
//Mapper class
public static class E_EMapper extends MapReduceBase implements
Mapper&lt;LongWritable ,/*Input key Type */ Text,            /*Input value Type*/
Text,            /*Output key Type*/ IntWritable&gt;       /*Output value Type*/
{
//Map function
public void map(LongWritable key, Text value, OutputCollector&lt;Text, IntWritable&gt; output,
Reporter reporter) throws IOException
{
String line = value.toString(); String lasttoken = null;
StringTokenizer s = new StringTokenizer(line,"\t");
String year = s.nextToken();
while(s.hasMoreTokens())
{
lasttoken=s.nextToken();
}
int avgprice = Integer.parseInt(lasttoken);
output.collect(new Text(year), new IntWritable(avgprice));
}
}
//Reducer class
public static class E_EReduce extends MapReduceBase implements
Reducer&lt; Text, IntWritable, Text, IntWritable &gt;
{
//Reduce function
public  void reduce( Text key, Iterator &lt;IntWritable&gt; values, OutputCollector&lt;Text,
IntWritable&gt; output, Reporter reporter) throws
IOException
{
int maxavg=30;
int val=Integer.MIN_VALUE;
while (values.hasNext())
{
if((val=values.next().get())&gt;maxavg)
{
output.collect(key, new IntWritable(val));
}
}
}
```

```
}
//Main function
public static void main(String args[])throws Exception
{
JobConf conf = new JobConf(ProcessUnits.class);
conf.setJobName("max_eletricityunits");
conf.setOutputKeyClass(Text.class);
conf.setOutputValueClass(IntWritable.class);
conf.setMapperClass(E_EMapper.class);
conf.setCombinerClass(E_EReduce.class);
conf.setReducerClass(E_EReduce.class);
conf.setInputFormat(TextInputFormat.class);
conf.setOutputFormat(TextOutputFormat.class);
FileInputFormat.setInputPaths(conf, new Path(args[0]));
FileOutputFormat.setOutputPath(conf, new Path(args[1]));
JobClient.runJob(conf);
}
```

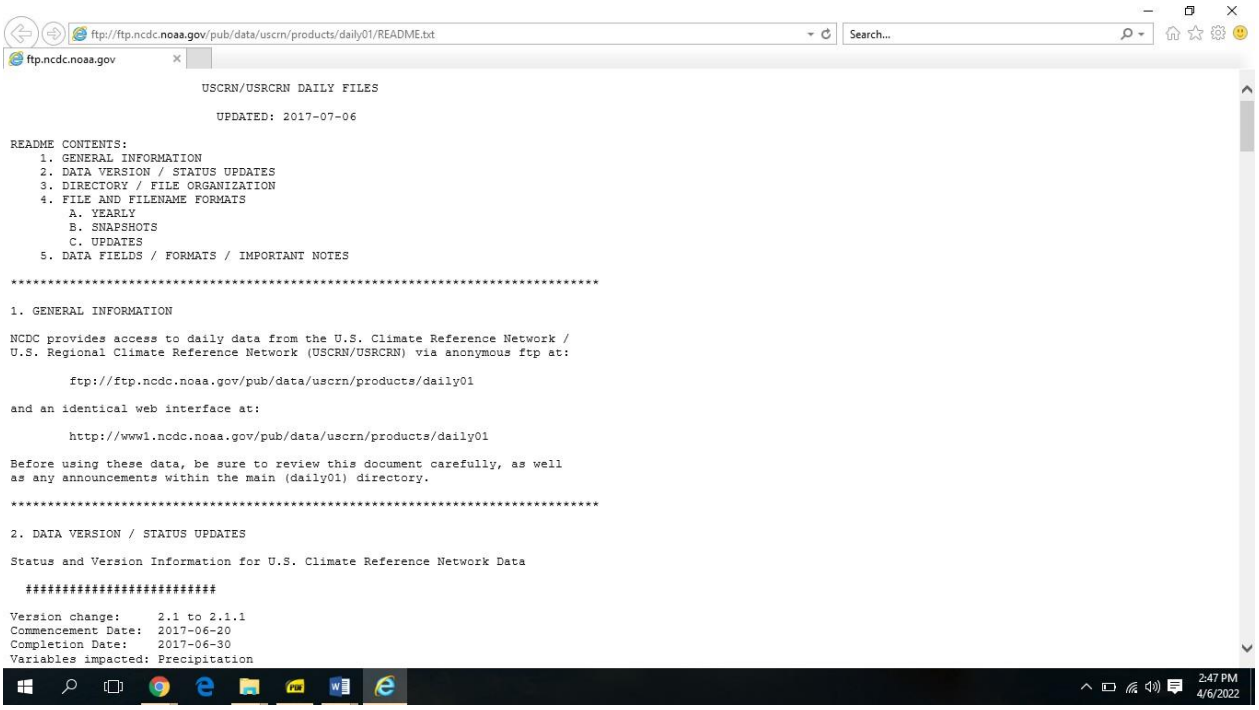
EXP NO: 7	MapReduce to analyze weather data set and print whether the day is shinny or cool
17/02/2026	

AIM: To Develop a MapReduce to analyze weather data set and print whether the day is shinny or cool day.

NOAA's National Climatic Data Center (***NCDC***) is responsible for preserving, monitoring, assessing, and providing public access to weather data.

NCDC provides access to daily data from the U.S. Climate Reference Network / U.S. Regional Climate Reference Network (USCRN/USRCRN) via anonymous ftp at:

Dataset ftp://ftp.ncdc.noaa.gov/pub/data/uscrn/products/daily01



```
                    USCRN/USRCRN DAILY FILES

                      UPDATED: 2017-07-06

README CONTENTS:
    1. GENERAL INFORMATION
    2. DATA VERSION / STATUS UPDATES
    3. DIRECTORY / FILE ORGANIZATION
    4. FILE AND FILENAME FORMATS
        A. YEARLY
        B. SNAPSHOTS
        C. UPDATES
    5. DATA FIELDS / FORMATS / IMPORTANT NOTES

*******************************************************************************

1. GENERAL INFORMATION

NCDC provides access to daily data from the U.S. Climate Reference Network /
U.S. Regional Climate Reference Network (USCRN/USRCRN) via anonymous ftp at:

        ftp://ftp.ncdc.noaa.gov/pub/data/uscrn/products/daily01

and an identical web interface at:

        http://www1.ncdc.noaa.gov/pub/data/uscrn/products/daily01

Before using these data, be sure to review this document carefully, as well
as any announcements within the main (daily01) directory.

*******************************************************************************

2. DATA VERSION / STATUS UPDATES

Status and Version Information for U.S. Climate Reference Network Data

  ##########################

Version change:     2.1 to 2.1.1
Commencement Date:  2017-06-20
Completion Date:    2017-06-30
Variables impacted: Precipitation
```

After going through wordcount mapreduce guide, you now have the basic idea of how a mapreduce program works. So, let us see a complex mapreduce program on weather dataset. Here I am using one of the dataset of year 2015 of Austin, Texas . We will do analytics on the dataset and classify whether it was a hot day or a cold day depending on the temperature recorded by NCDC.

NCDC gives us all the weather data we need for this

mapreduce project. The dataset which we will be using looks

like below snapshot.

ftp://ftp.ncdc.noaa.gov/pub/data/uscrn/products/daily01/2015/CRND0103-2015- TX_Austin_33_NW.txt

ftp://ftp.ncdc.noaa.gov/pub/data/uscrn/products/daily01/2015/CRND0103-2015-TX_Austin_33_NW.txt

```
23907 20150101  2.423  -98.08   30.62     2.2    -0.6     0.8     0.9     6.2     1.47 C     3.7     1.1
2.5    99.9    85.4    97.2   0.369   0.308 -99.000 -99.000 -99.000     7.0     8.1 -9999.0 -9999.0 -9999.0
23907 20150102  2.423  -98.08   30.62     3.5     1.3     2.4     2.2     9.0     1.43 C     4.9     2.3
3.1   100.0    98.8    99.8   0.391   0.327 -99.000 -99.000 -99.000     7.1     7.9 -9999.0 -9999.0 -9999.0
23907 20150103  2.423  -98.08   30.62    15.9     2.3     9.1     7.5     2.9    11.00 C    16.4     2.9
7.3   100.0    34.8    73.7   0.450   0.397 -99.000 -99.000 -99.000     7.6     7.9 -9999.0 -9999.0 -9999.0
23907 20150104  2.423  -98.08   30.62     9.2    -1.3     3.9     4.2     0.0    13.24 C    12.4    -0.5
4.9    82.0    40.6    61.7   0.414   0.352 -99.000 -99.000 -99.000     7.3     7.9 -9999.0 -9999.0 -9999.0
23907 20150105  2.423  -98.08   30.62    10.9    -3.7     3.6     2.6     0.0    13.37 C    14.7    -3.0
3.8    77.9    33.3    57.4   0.399   0.340 -99.000 -99.000 -99.000     6.3     7.0 -9999.0 -9999.0 -9999.0
23907 20150106  2.423  -98.08   30.62    20.2     2.9    11.6    10.9     0.0    12.90 C    22.0     1.6
9.9    67.7    30.2    49.3   0.395   0.335 -99.000 -99.000 -99.000     8.0     8.0 -9999.0 -9999.0 -9999.0
23907 20150107  2.423  -98.08   30.62    10.9    -3.4     3.8     4.5     0.0    12.68 C    12.4    -2.1
5.5    82.7    36.5    55.7   0.387   0.328 -99.000 -99.000 -99.000     7.6     8.3 -9999.0 -9999.0 -9999.0
23907 20150108  2.423  -98.08   30.62     0.6    -7.9    -3.6    -3.3     0.0     4.98 C     3.9    -4.8
-0.5    57.7    37.6    48.1   0.372   0.316 -99.000 -99.000 -99.000     4.7     6.1 -9999.0 -9999.0 -9999.0
23907 20150109  2.423  -98.08   30.62     2.0     0.1     1.0     0.8     0.0     2.52 C     4.1     1.2
2.5    87.8    48.9    64.4   0.368   0.312 -99.000 -99.000 -99.000     5.4     6.2 -9999.0 -9999.0 -9999.0
23907 20150110  2.423  -98.08   30.62     0.5    -2.0    -0.8    -0.6     3.3     2.11 C     2.5    -0.1
```

Step 1

Download the complete project using below link.

https://drive.google.com/file/d/0B2SFMPvhXPQ5bUdoVFZsQjE2ZDA/view?usp=sharing

```
import java.io.IOException;
import java.util.Iterator;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;

import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

import  org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;
```

```
import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.conf.Configuration;
public class MyMaxMin {
public static class MaxTemperatureMapper extends
Mapper<LongWritable, Text, Text, Text> {
  /**
 * @method map
* This method takes the input as text data type
* Now leaving the first five tokens,it takes 6th token is taken as temp_max and
* 7th token is taken as temp_min. Now  temp_max > 35  and
temp_min < 10 are passed to the reducer.
*/ @Override
public void map(LongWritable arg0, Text Value, Context 2 context) throws IOException,
InterruptedException {
 //Converting the record (single line) to String and storing it in a String variable line
String line = Value.toString();
//Checking if the line is not empty
  if (!(line.length() == 0)) {
  //date
String date = line.substring(6, 14);
//maximum temperature
float temp_Max = Float
        parseFloat(line.substring(39, 45).trim());
//minimum temperature
float temp_Min = Float
parseFloat(line.substring(47, 53).trim());
//if maximum temperature is greater than 35 , its a hot day
```

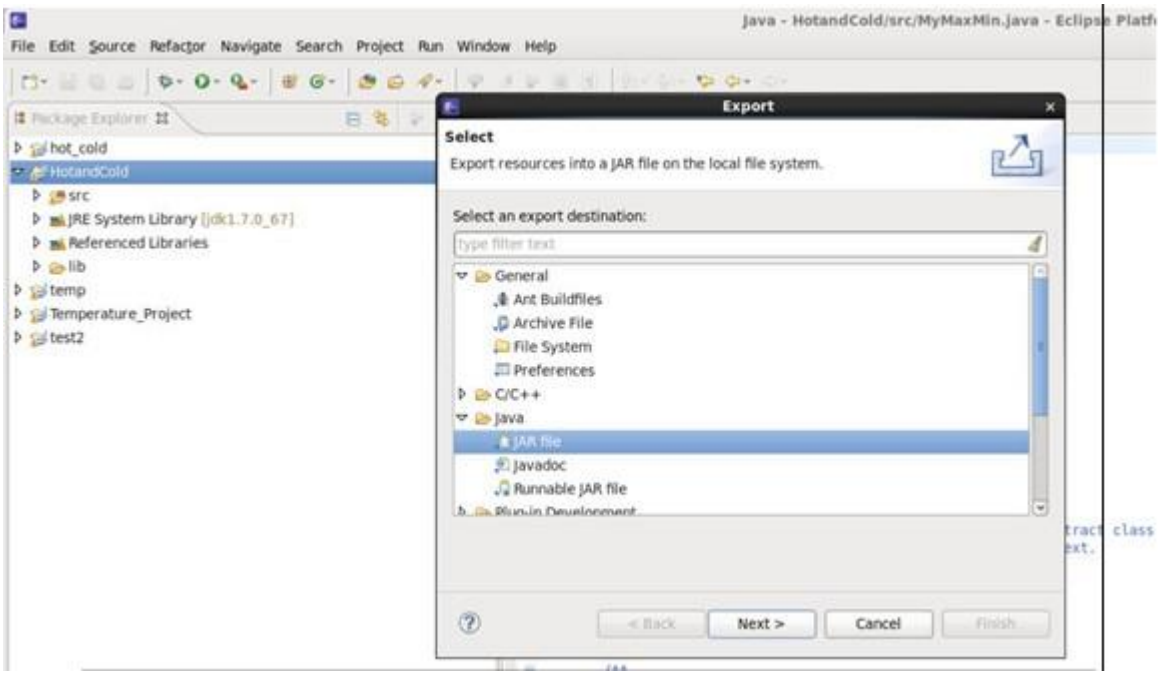
```
if (temp_Max > 35.0) {
// Hot day
context.write(new Text("Hot Day " + date),
new Text(String.valueOf(temp_Max)));
}
//if minimum temperature is less than 10, it's a cold day

if (temp_Min < 10) {

        // Cold day

context.write(new Text("Cold Day " + date),

new  Text(String.valueOf(temp_Min)));
}
}
}
}
//Reducer
*MaxTemperatureReducer class is static and extends Reducer abstract

having four hadoop generics type Text, Text, Text, Text.

*/

public static class MaxTemperatureReducer extends Reducer<Text, Text, Text,
Text> {

public void reduce (Text Key, Iterator<Text> Values, Context context) throws
IOException, Interrupted Exception {
String temperature = Values.next().toString();
context.write(Key, new Text(temperature));
}
}
public static void main(String[] args) throws Exception {
Configuration conf = new Configuration();
```

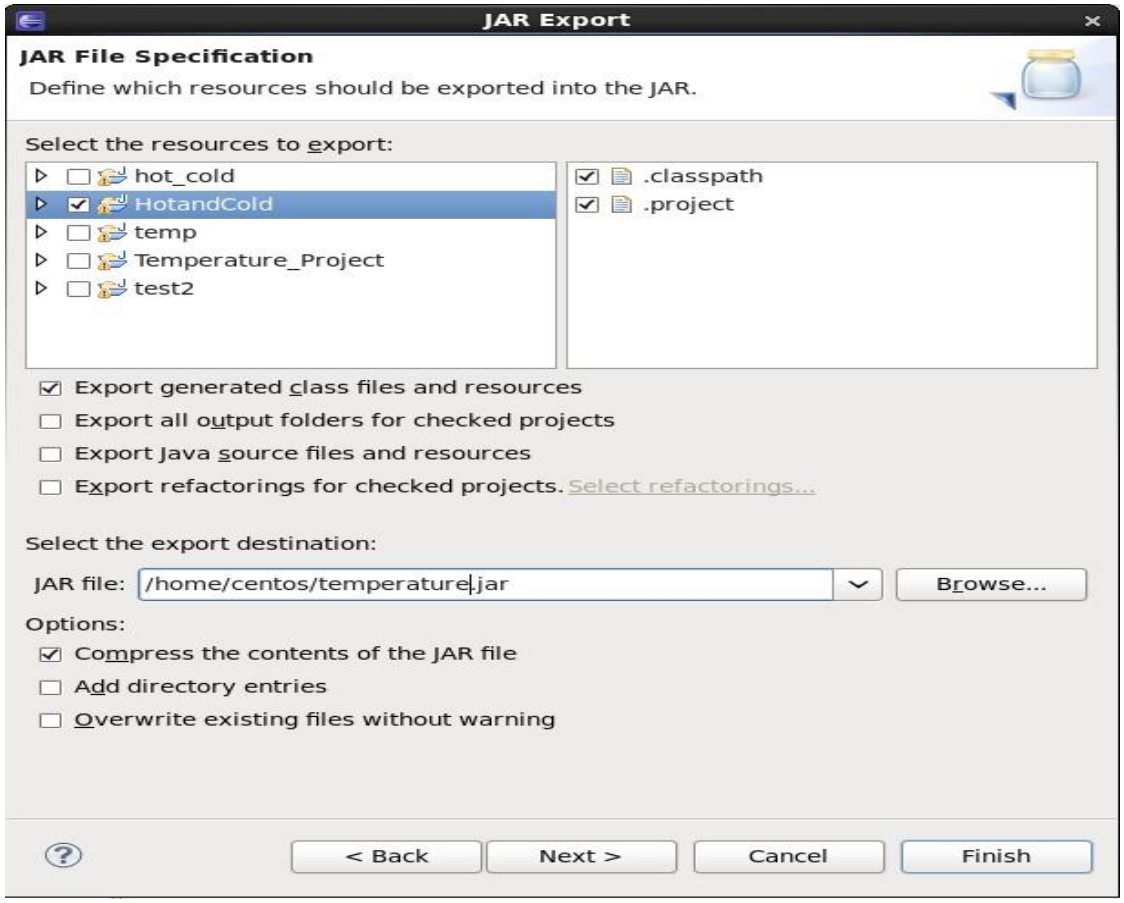
```
Job job = new Job(conf, "weather example");

job.setJarByClass(MyMaxMin.class);

job.setMapOutputKeyClass(Text.class);

job.setMapOutputValueClass(Text.class);
job.setMapperClass(MaxTemperatureMapper.class);

job.setReducerClass(MaxTemperatureReducer.class);

job.setInputFormatClass(TextInputFormat.class);

job.setOutputFormatClass(TextOutputFormat.class);

Path OutputPath = new Path(args[1]);

FileInputFormat.addInputPath(job, new Path(args[0]));

FileOutputFormat.setOutputPath(job, new Path(args[1]));

OutputPath.getFileSystem(conf).delete(OutputPath);

System.exit(job.waitForCompletion(true) ? 0 : 1);

}

}
```

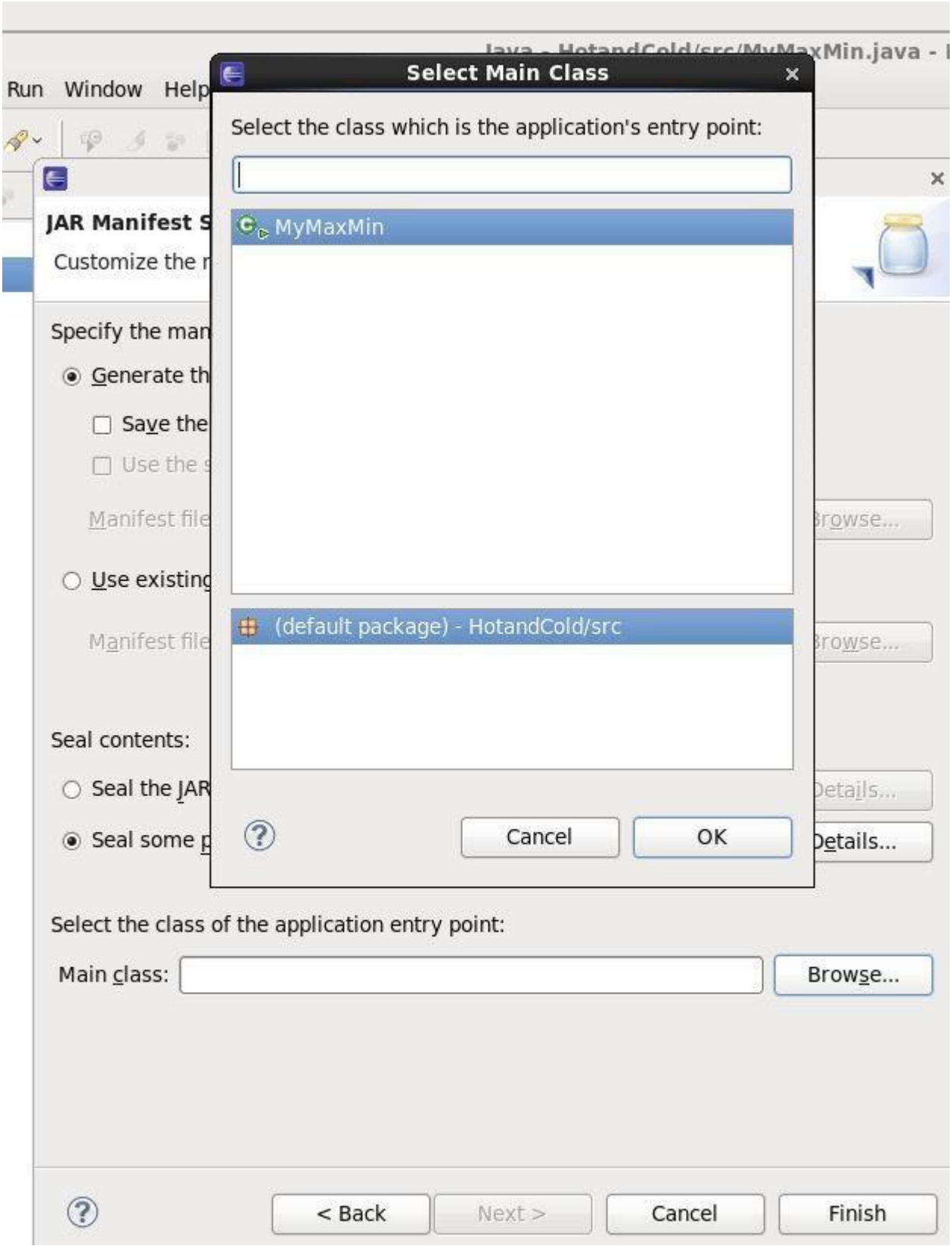
Import the project in eclipse IDE in the same way it was told in earlier guide and change the jar paths with the jar files present in the lib directory of this project.

When the project is not having any error, we will export it as a jar file, same as we did in wordcount mapreduce guide. Right Click on the Project file and click on Export. Select jar file.

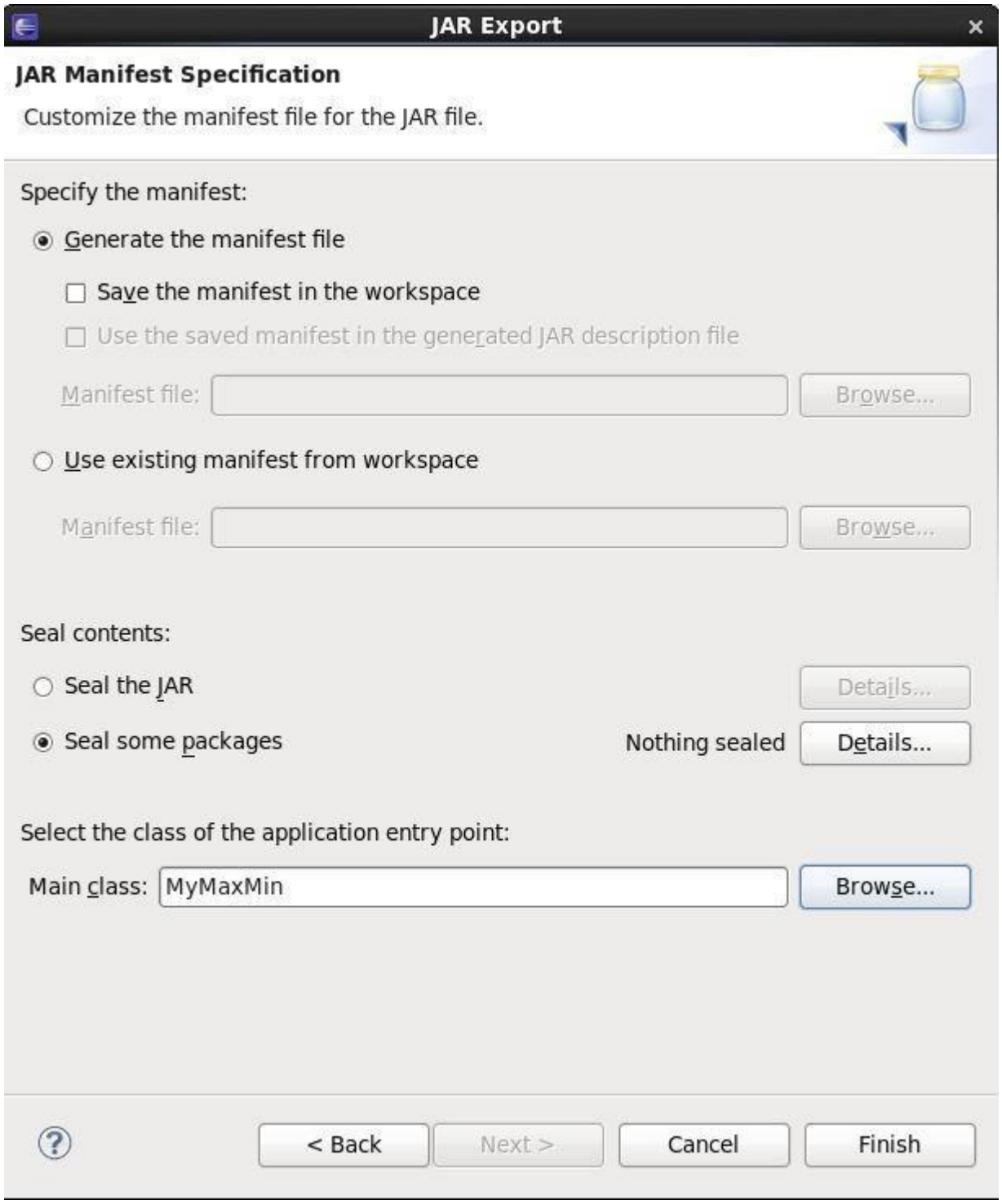
Give the path where you want to save the file.


Java - HotandCold/src/MyMaxMin.java - Eclipse Platf
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
hot_cold
HotandCold
src
JRE System Library [jdk1.7.0_67]
Referenced Libraries
lib
temp
Temperature_Project
test2
Export
Select
Export resources into a JAR file on the local file system.
Select an export destination:
type filter text
General
Ant Buildfiles
Archive File
File System
Preferences
C/C++
Java
JAR file
Javadoc
Runnable JAR file
< Back
Next >
Cancel
Finish


JAR Export
JAR File Specification
Define which resources should be exported into the JAR.
Select the resources to export:
hot_cold
HotandCold
temp
Temperature_Project
test2
.classpath
.project
Export generated class files and resources
Export all output folders for checked projects
Export Java source files and resources
Export refactorings for checked projects. Select refactorings...
Select the export destination:
JAR file: /home/centos/temperature.jar
Browse...
Options:
Compress the contents of the JAR file
Add directory entries
Overwrite existing files without warning
< Back
Next >
Cancel
Finish


Select Main Class
Select the class which is the application's entry point:
MyMaxMin
(default package) - HotandCold/src
Cancel
OK
Run
Window
Help
JAR Manifest S
Customize the r
Specify the man
Generate th
Save the
Use the s
Manifest file
Browse...
Use existing
Manifest file
Browse...
Seal contents:
Seal the JAR
Details...
Seal some p
Details...
Select the class of the application entry point:
Main class:
Browse...
< Back
Next >
Cancel
Finish

Click on Finish to export.



You can download the jar file directly using below link

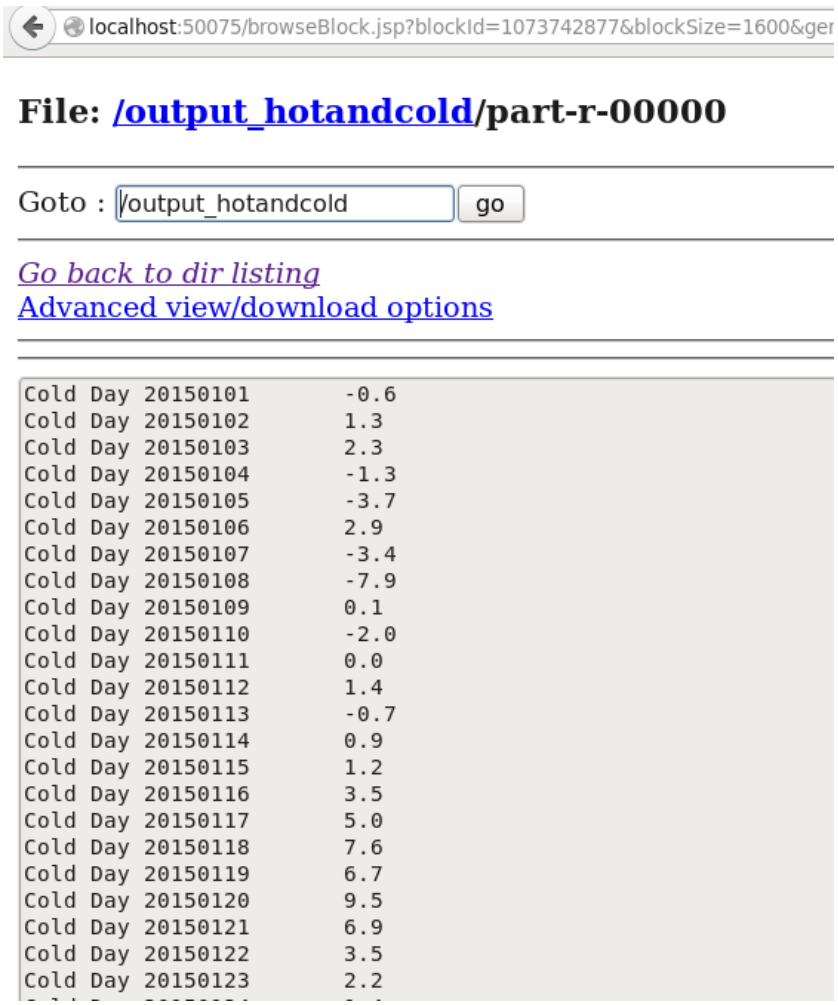
temperature.jar

https://drive.google.com/file/d/0B2SFMPvhXPQ5RUlZZDZSR3FYVDA/view?usp=sharing

Download Dataset used by me using below

link weather_data.txt

https://drive.google.com/file/d/0B2SFMPvhXPQ5aFVILXAxbFh6ejA/view?usp=sharing



EXP NO: 8	Program to calculate the maximum recorded temperature by year wise for the weather dataset in Pig Latin
24/02/2026	

AIM: To Develop a program to calculate the maximum recorded temperature by year wise for the weather dataset in Pig Latin

Description:

The National Climatic Data Center (NCDC) is the world's largest active archive of weather data. I downloaded the NCDC data for year 1930 and loaded it in HDFS system. I implemented MapReduce program and Pig, Hove scripts to findd the Min, Max, avg temparature for diffrent stations.

Compiled the Java File: javac -classpath /home/student3/hadoop-common-2.6.1.jar:/home/student3/hadoop-mapreduce-client-core-2.6.1.jar:/home/student3/commons-cli-2.0.jar -d . MaxTemperature.java MaxTemperatureMapper.java MaxTemperatureReducer.java

Created the JAR file: jar -cvf hadoop-project.jar *class

Executed the jar file: hadoop jar hadoop-project.jar MaxTemperature /home/student3/Project/ /home/student3/Project_output111

Copy the output file to local hdfs dfs -copyToLocal /home/student3/Project_output111/part-r-00000

PIG Script

Pig -x local grunt> records = LOAD '/home/student3/Project/Project_Output/output111.txt' AS (year:chararray, temperature:int); grunt> DUMP records; grunt> grouped_records = GROUP records BY year; grunt> DUMP grouped_records; grunt> max_temp = FOREACH grouped_records GENERATE group,

Hive Script

Commands to create table in hive and to find average temperature

DROP TABLE IF EXISTS w_hd9467;

CREATE TABLE w_hd9467(year STRING, temperature INT) ROW FORMAT DELIMITED FIELDS TERMINATED BY ‘\t’;

LOAD DATA LOCAL INPATH '/home/student3/Project/Project_Output/output1.txt'

OVERWRITE INTO TABLE w_hd9467;

SELECT count(*) from w_hd9467;

SELECT * from w_hd9467 limit 5;

Query to find average temperature SELECT year, AVG(temperature) FROM w_hd9467 GROUP BY year;

MaxTemperature.java

```
import  org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MaxTemperature {

  public static void main(String[] args) throws Exception {
    if (args.length != 2) {
      System.err.println("Usage: MaxTemperature <input path> <output path>");
      System.exit(-1);
    }

    Job job = new Job();
    job.setJarByClass(MaxTemperature.class);
    job.setJobName("Max temperature");

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    job.setMapperClass(MaxTemperatureMapper.class);
    job.setReducerClass(MaxTemperatureReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
}
```

MaxTemperatureMapper.java

```
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class MaxTemperatureMapper
  extends Mapper<LongWritable, Text, Text, IntWritable> {

  private static final int MISSING = 9999;

  @Override
  public void map(LongWritable key, Text value, Context context)
      throws IOException, InterruptedException {

    String line = value.toString();
    String year = line.substring(15, 19);
    int airTemperature;
    if (line.charAt(87) == '+') { // parseInt doesn't like leading plus signs
      airTemperature = Integer.parseInt(line.substring(88, 92));
    } else {
      airTemperature = Integer.parseInt(line.substring(87, 92));
    }
    String quality = line.substring(92, 93);
    if (airTemperature != MISSING && quality.matches("[01459]")) {
      context.write(new Text(year), new IntWritable(airTemperature));
    }
  }
}
```

MaxTemperatureReducer.java

```
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class MaxTemperatureReducer
```

```
  extends Reducer<Text, IntWritable, Text, IntWritable> {

  @Override
  public void reduce(Text key, Iterable<IntWritable> values,
      Context context)
      throws IOException, InterruptedException {

    int maxValue = Integer.MIN_VALUE;
    for (IntWritable value : values) {
    context.write(key, value);
     // maxValue = Math.max(maxValue, value.get());
    }
    //context.write(key, new IntWritable(maxValue));
  }
}
```

EXP NO: 9	Queries to sort and aggregate the data in a table using HiveQL
03/03/2026	

AIM: To Write queries to sort and aggregate the data in a table using HiveQL

Description:

Hive is an open-source data warehousing solution built on top of Hadoop. It supports an SQL-like query language called HiveQL. These queries are compiled into MapReduce jobs that are executed on Hadoop. While Hive uses Hadoop for execution of queries, it reduces the effort that goes into writing and maintaining MapReduce jobs.

Hive supports database concepts like tables, columns, rows and partitions. Both primitive (integer, float, string) and complex data-types(map, list, struct) are supported. Moreover, these types can be composed to support structures of arbitrary complexity. The tables are serialized/deserialized using default serializers/deserializer. Any new data format and type can be supported by implementing SerDe and ObjectInspector java interface.

HiveQL - ORDER BY and SORT BY Clause

By using HiveQL ORDER BY and SORT BY clause, we can apply sort on the column. It returns the result set either in ascending or descending order. Here, we are going to execute these clauses on the records of the below table:

emp

Id	Name	Salary	Department
1	Gaurav	30000	Developer
2	Aryan	20000	Manager
3	Vishal	40000	Manager
4	John	10000	Trainer
5	Henry	25000	Developer
6	William	9000	Developer
7	Lisa	25000	Manager
8	Ronit	20000	Trainer

HiveQL - ORDER BY Clause

In HiveQL, ORDER BY clause performs a complete ordering of the query result set. Hence, the complete data is passed through a single reducer. This may take much time in the execution of large datasets. However, we can use LIMIT to minimize the sorting time.

Example:

Select the database in which we want to create a table.

hive> use hiveql;

```
codegyani@ubuntu64server: ~
hive> use hiveql;
OK
Time taken: 0.067 seconds
hive>
```

Now, create a table by using the following command:

hive> create table emp (Id int, Name string , Salary float, Department string)
row format delimited
fields terminated by ',' ;

```
codegyani@ubuntu64server: ~
hive> create table emp (Id int, Name string , Salary float, Department string)

    > row format delimited
    > fields terminated by ',' ;
OK
Time taken: 0.419 seconds
hive>
```

Load the data into the table

hive> load data local inpath '/home/codegyani/hive/emp_data' into table emp;

```
codegyani@ubuntu64server: ~
hive> load data local inpath '/home/codegyani/hive/emp_data' into table emp;
Loading data to table hiveql.emp
Table hiveql.emp stats: [numFiles=1, totalSize=200]
OK
Time taken: 1.411 seconds
hive>
```

Now, fetch the data in the descending order by using the following command

hive> select * from emp order by salary desc;

```
codegyani@ubuntu64server: ~
hive> select * from emp order by salary desc;
Query ID = codegyani_20190802063522_65b28a82-4d0b-492a-ae25-2faef1471b65
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1555046592674_0032, Tracking URL = http://ubuntu64server:8088
/proxy/application_1555046592674_0032/
Kill Command = /home/codegyani/hadoop-2.7.1//bin/hadoop job  -kill job_155504659
2674_0032
```

```
codegyani@ubuntu64server: ~
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2019-08-02 06:36:48,908 Stage-1 map = 0%,  reduce = 0%
2019-08-02 06:37:49,829 Stage-1 map = 0%,  reduce = 0%
2019-08-02 06:38:02,548 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 7.03 se
c
2019-08-02 06:39:03,090 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 10.31 s
ec
2019-08-02 06:39:18,347 Stage-1 map = 100%,  reduce = 67%, Cumulative CPU 12.98
sec
2019-08-02 06:39:35,537 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 22.34
 sec
MapReduce Total cumulative CPU time: 22 seconds 340 msec
Ended Job = job_1555046592674_0032
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 22.34 sec   HDFS Read: 6788 H
DFS Write: 227 SUCCESS
```

```
codegyani@ubuntu64server: ~
Total MapReduce CPU Time Spent: 22 seconds 340 msec
OK
3       "Vishal"        40000.0 Manager
1       "Gaurav"        30000.0 Developer
7       "Lisa"  25000.0 Manager
5       "Henry" 25000.0 Developer
8       "Ronit" 20000.0 Trainer
2       "Aryan" 20000.0 Manager
4       "John"  10000.0 Trainer
6       "William"       9000.0  Developer
NULL    NULL    NULL    NULL
Time taken: 257.304 seconds, Fetched: 9 row(s)
hive>
```

HiveQL - SORT BY Clause

The HiveQL SORT BY clause is an alternative of ORDER BY clause. It orders the data within each reducer. Hence, it performs the local ordering, where each reducer's output is sorted separately. It may also give a partially ordered result.

Example:

Let's fetch the data in the descending order by using the following command

hive> select * from emp sort by salary desc;

```
codegyani@ubuntu64server: ~
hive> select * from emp sort by salary desc;
Query ID = codegyani_20190802065014_f877314f-8d92-428f-8a9f-1b6a9b67c328
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1555046592674_0033, Tracking URL = http://ubuntu64server:8088
/proxy/application_1555046592674_0033/
Kill Command = /home/codegyani/hadoop-2.7.1//bin/hadoop job  -kill job_155504659
2674_0033
```

```
codegyani@ubuntu64server: ~
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2019-08-02 06:51:51,188 Stage-1 map = 0%,  reduce = 0%
2019-08-02 06:52:51,366 Stage-1 map = 0%,  reduce = 0%
2019-08-02 06:53:12,252 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 10.78 s
ec
2019-08-02 06:54:13,249 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 10.78 s
ec
2019-08-02 06:54:23,926 Stage-1 map = 100%,  reduce = 67%, Cumulative CPU 15.14
sec
2019-08-02 06:54:37,633 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 22.69
 sec
MapReduce Total cumulative CPU time: 22 seconds 690 msec
Ended Job = job_1555046592674_0033
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 22.69 sec   HDFS Read: 6788 H
DFS Write: 227 SUCCESS
```

```
codegyani@ubuntu64server: ~
Total MapReduce CPU Time Spent: 22 seconds 690 msec
OK
3       "Vishal"        40000.0 Manager
1       "Gaurav"        30000.0 Developer
7       "Lisa"  25000.0 Manager
5       "Henry" 25000.0 Developer
8       "Ronit" 20000.0 Trainer
2       "Aryan" 20000.0 Manager
4       "John"  10000.0 Trainer
6       "William"       9000.0  Developer
NULL    NULL    NULL    NULL
Time taken: 268.62 seconds, Fetched: 9 row(s)
```

Cluster By:

Cluster By used as an alternative for both Distribute BY and Sort BY clauses in Hive-QL.

Cluster BY clause used on tables present in Hive. Hive uses the columns in Cluster by to distribute the rows among reducers. Cluster BY columns will go to the multiple reducers.

- It ensures sorting orders of values present in multiple reducers

For example, Cluster By clause mentioned on the Id column name of the table employees_guru table. The output when executing this query will give results to multiple reducers at the back end. But as front end it is an alternative clause for both Sort By and Distribute By.

Example:
SELECT Id, Name from employees_guru CLUSTER BY Id;

```
hive> Select Id,Name from employees guru CLUSTER BY Id;
Query ID = hd[illegible]er_20151105165000_72cedc06-a797-48b1-a120-
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not [illegible] from inpu
In order to change the ave[illegible] (in byt
  set hive.exec.reducers.b[illegible] >
In order to limit the maxi[illegible]
  set hive.exec.reducers.m[illegible]
In order to set a constant [illegible]
  set mapred.reduce.tasks=<number>
Starting Job = job_201511051442_0009, Tracking URL = http
Kill Command = /usr/local/hadoop-1.2.1/libexec/../bin/had
Hadoop job information for Stage-1: number of mappers: 1;
2015-11-05 16:50:08,541 Stage-1 map = 0%,  reduce = 0%
2015-11-05 16:50:10,546 [illegible] reduce = 0%,
2015-11-05 16:50:17,563 [illegible] reduce = 100
MapReduce Total cumulat[illegible]nds 600 msec
Ended Job = job_20151[illegible]
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 1.6 se
Total MapRe[illegible]e CPU Time Spent: 1 seconds 600 msec
OK
101     Rajesh
102     Rajiv
103     Animesh
104     Anirudh
105     Santosh
106     Ramesh
107     Sravanthi
108     Sravan
109     Suresh
110     Ravi
111     Syam
Time taken: 18.941 seconds, Fetched: 11 row(s)
```

1

cluster by query

cluster by query output

2

EXP NO: 10	Java application to find the maximum temperature using Spark
10/03/2026	

AIM: To Develop a Java application to find the maximum temperature using Spark.

Sourcecode:

```
import org.apache.spark._;
object testfilter extends App {
val conf=new SparkConf().setMaster("local[2]").setAppName("testfilter")
val sc = new SparkContext(conf)
System.setProperty("hadoop.home.dir", "c://winutil//")
val input=sc.textFile("file:///D://sparkprog//temp//stats.txt")
val line=input.map(x=>x.split("\t
val city=line.map(x=>(x(3)+"\t x(4)))
val rdd3=city.map(x=>x.split("\t
val maintemp=rdd3.map(x=>((x(0),x(1))))
val grp= maintemp.groupByKey()
val main = grp.map{case (x,iter) => (x,iter.toList.max)}
for ( i<- main)
{
print(i)
print("\n")
}
}
```

EXP NO: 11	HIVE OPERATIONS
17/03/2026	

AIM: To Use Hive to create, alter, and drop databases, tables, views, functions, and indexes.

RESOURCES:
VMWare, XAMPP Server, Web Browser, 1GB RAM, Hard Disk 80 GB.

PROGRAM LOGIC:
SYNTAX for HIVE Database Operations
DATABASE Creation
CREATE DATABASE|SCHEMA [IF NOT EXISTS] <database name>
Drop Database Statement
DROP DATABASE StatementDROP (DATABASE|SCHEMA) [IF EXISTS]
database_name [RESTRICT|CASCADE];
Creating and Dropping Table in HIVE
CREATE [TEMPORARY] [EXTERNAL] TABLE [IF NOT EXISTS] [db_name.]
table_name
[(col_name data_type [COMMENT col_comment], ...)]
[COMMENT table_comment] [ROW FORMAT row_format] [STORED AS
file_format]
Loading Data into table log_data
Syntax:
LOAD DATA LOCAL INPATH '<path>/u.data' OVERWRITE INTO TABLE
u_data;
Alter Table in HIVE
Syntax
ALTER TABLE name RENAME TO new_name
ALTER TABLE name ADD COLUMNS (col_spec[, col_spec ...])
ALTER TABLE name DROP [COLUMN] column_name
ALTER TABLE name CHANGE column_name new_name new_type
ALTER TABLE name REPLACE COLUMNS (col_spec[, col_spec ...])
Creating and Dropping View
CREATE VIEW [IF NOT EXISTS] view_name [(column_name [COMMENT
column_comment], ...)] [COMMENT table_comment] AS SELECT ...
Dropping View
Syntax:
DROP VIEW view_name
Functions in HIVE
String Functions:- round(), ceil(), substr(), upper(), reg_exp() etc

Date and Time Functions:- year(), month(), day(), to_date() etc
Aggregate Functions :- sum(), min(), max(), count(), avg() etc
INDEXES
CREATE INDEX index_name ON TABLE base_table_name (col_name, ...)
AS 'index.handler.class.name'
[WITH DEFERRED REBUILD]
[IDXPROPERTIES (property_name=property_value, ...)]
[IN TABLE index_table_name]
[PARTITIONED BY (col_name, ...)]
[
[ROW FORMAT ...] STORED AS ...
| STORED BY ...
]
[LOCATION hdfs_path]
[TBLPROPERTIES (...)]
Creating Index
CREATE INDEX index_ip ON TABLE log_data(ip_address) AS
'org.apache.hadoop.hive.ql.index.compact.CompactIndexHandler' WITH DEFERRED
REBUILD;
Altering and Inserting Index
ALTER INDEX index_ip_address ON log_data REBUILD;
Storing Index Data in Metastore
SET
hive.index.compact.file=/home/administrator/Desktop/big/metastore_db/tmp/index_ipadd
ress_result;
SET
hive.input.format=org.apache.hadoop.hive.ql.index.compact.HiveCompactIndexInputFor
mat;
Dropping Index
DROP INDEX INDEX_NAME on TABLE_NAME;

RUBRICS EVALUATION

Performance Criteria	Scale 1 (0-25%)	Scale 2 (26-50%)	Scale 3 (51-75%)	Scale 4 (76-100%)	Score (Numerical)
Understandability Ability to analyses Problem and Identify solution	Unable to understandthe problem.	Able to understand the problem partially and unable to identify the solution	Able to understand the problem completely but unable to identify the solution	Able to understand the problem completely and able to provide alternative solution too.	
Logic Ability to specify Conditions & control flow that are appropriatefor the problem domain.	Program logic is incorrect	Program logic is on the right track but has several errors	Program logic is mostly correct, but may contain an occasional boundary error or redundant or contradictory condition.	Program logic is correct, with no known boundary errors, and no redundant or contradictory conditions.	
Debugging Ability to execute /debug	Unable to execute program	Unable to debug several errors.	Able to execute program with several warnings.	Able to execute program completely	
Correctness Ability to code formulaeand algorithms that reliably produce correct answers or appropriate results.	Program does not produce correct answers or appropriate results for most inputs.	Program approaches correct answers or appropriate results for most inputs, but can contain miscalculations in some cases.	Program produces correct answers or appropriate results for most inputs.	Program produces correct answers or appropriate results for all inputs tested.	
Completeness Ability to demonstrateand deliver on time.	Unable to explain the code and the code was overdue.	Unable to explain the code and the code submission was late.	Able to explain code and the program was delivered within the due date.	Able to explain code and the program was delivered on time.	
				TOTAL	

Sign